

Microsoft Excel

Course Project & Reference Guide

JTS English Medium School

Student: **Rehan Hassan**

Class: **7**

Roll No: **2**

1 Introduction to Excel

Microsoft Excel is a spreadsheet program developed by Microsoft. It is used to organize, calculate, analyze and present data in the form of rows and columns. Excel provides formulas and functions that make calculations faster and more accurate. It can be used to prepare tables, marksheets, budgets, reports and charts. Excel is widely used in schools, offices, businesses and other organizations for managing and analyzing data efficiently.

2 Mathematical Functions in Excel

Mathematical functions are built-in Excel functions used to perform calculations on numbers and ranges of cells. A formula normally begins with an equal sign (=). Some commonly used mathematical functions are shown below:

FUNCTION	PURPOSE	EXAMPLE	OUTPUT
SUM(range)	Returns the sum of values in a range.	=SUM(4,5,6)	15
PRODUCT(range)	Multiplies the values in a range.	=PRODUCT(8,2,4)	64
AVERAGE(range)	Finds the average of the given numbers.	=AVERAGE(7,8,5)	6.6667
INT(number)	Rounds a number down to an integer.	=INT(7.15)	7
ROUND(num, digits)	Rounds a number to the specified digits.	=ROUND(56,2)	56
SQRT(number)	Returns the square root of a number.	=SQRT(25)	5
POWER(num, power)	Returns a number raised to a power.	=POWER(6,2)	36
MOD(num, divisor)	Returns the remainder after division.	=MOD(9,2)	1
COUNT(range)	Counts the number of values in a range.	=COUNT(2,4,5,6)	4
MAX(range)	Finds the largest value in a range.	=MAX(8,5,6,9)	9
MIN(range)	Finds the smallest value in a range.	=MIN(18,5,26,9)	5

3 Text Functions in Excel

Text functions are used to perform operations on text or strings. They can join text, extract characters, count characters, and change the case of text. These functions are useful when working with names, labels, and other textual data.

FUNCTION	PURPOSE	EXAMPLE	OUTPUT
CONCAT(t1, t2)	Joins two or more text strings.	<code>=CONCAT("Key", "Board")</code>	KeyBoard
LEFT(text, num)	Returns characters from the left side of a string.	<code>=LEFT("Keyboard", 3)</code>	Key
RIGHT(text, num)	Returns characters from the right side of a string.	<code>=RIGHT("Keyboard", 5)</code>	board
LEN(text)	Returns the length of a text string.	<code>=LEN("Keyboard")</code>	8
UPPER(text)	Converts text into uppercase.	<code>=UPPER("Keyboard")</code>	KEYBOARD
LOWER(text)	Converts text into lowercase.	<code>=LOWER("Keyboard")</code>	keyboard

4 Minimum Balance

Minimum balance refers to the minimum amount of money that a customer is required to maintain in a bank account. Banks may set different minimum balance requirements depending on the type of account and the services provided. If the balance falls below the required amount, the bank may charge a penalty or service fee according to its rules. However, some basic savings and zero-balance accounts may not require customers to maintain a fixed minimum balance. Therefore, customers should check the bank's minimum balance requirements and applicable charges before opening an account.

5 Conclusion

Microsoft Excel is a useful and powerful spreadsheet application for organizing and working with data. Mathematical functions help perform calculations quickly, while text functions make it easier to manage and modify textual information. Understanding functions such as SUM, AVERAGE, PRODUCT, MIN, MAX, CONCAT, LEFT, RIGHT, LEN, UPPER, and LOWER helps users work more efficiently in Excel. Learning these basic features provides a strong foundation for using Excel in schoolwork, projects, and everyday data management.